

Where the Rent Moves Next: Four Binding Constraints Capital Has Not Yet Found

The shortage everyone funds is at the factory. The rent has already moved on, to the deck of the ship, the certified hands, and the isotope nobody in the West can legally make.

The frame: when a system scales, the money does not stay where the headlines are. It moves to the one input that cannot scale with it. Our job is to name that input, and the date it starts to bite, before the market prices it in.

Area: any area, wide open across all industries • Horizon: 2030 to 2040

How we work: cast wide for ideas, then test every one hard against the price and the supply chain. We name the specific input, not the theme.

Two numbers per call: how strong we think the case is, and the odds on the exact, dated version, which is the one we are graded on.

Status: the calls that survived our own attempt to break them. Drafted 2026-06-15.

The board: 4 calls that survived the cut

The cross-cutting read

We see the same shift play out across all four calls, in four different physical systems. Capital and headlines crowd into the fast, fundable, capital-intensive layer (cable plants, copper mines, fusion magnets, grid transformers). Once that layer clears, the binding constraint does not disappear. It moves one node downstream onto something money cannot mint on the same clock. In three of the four cases that residual node is people or a permitted process, not steel. Certified offshore HV jointers, settings-grade protection-and-control relay engineers, journeyman linemen: a multi-year, retiring, non-compressible apprenticeship pipeline sits under every cable, substation, and interconnection. In copper the residual is qualified non-China smelting capacity, irreversible multi-billion-dollar assets being culled asymmetrically while the market still trades a mine-and-demand story. The Li-6 fusion call is the cleanest version of the pattern: the input is physically required, its only scaled production route is treaty-banned, and live capacity sits with two adversary states. The price channel is the giveaway every time. The market is paying for the hardware layer, while the genuinely inelastic node has no liquid market anyone trades, which is exactly why we think it is the live edge. The honest tax, the reason the probability on the exact dated version sits well below our conviction in each thesis, is that these are conjunctive, dated calls and the downstream node always has one elasticity leak: offshored relay-settings work, robotic jointing, new state-backed Asian smelters, or a natural-lithium blanket redesign.

At a glance

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
P1	We think the copper debate is aimed at the wrong end of the chain. The market fixates on the mine: too few new deposits, decade-long permitting, an 80-percent-of-needs-by-2030 gap. That is real, it is also consensus, and it is priced. The quieter fact is that China runs about 97 percent of global copper smelting and refining capacity, and the economics of conversion now punish everyone else. The treatment and refining charge, the fee a smelter earns to turn concentrate into metal, settled at zero per tonne on the 2026	Qualified non-China copper smelting and refining capacity. The dated catalyst is the zero-to-negative treatment-charge r	72%	55%	2033-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	annual benchmark, the lowest ever agreed, and the spot charge has gone deeply negative, hitting roughly minus 78 dollars a tonne by April 2026. A negative charge means a smelter pays for the right to operate. Chinese smelters survive on byproduct gold, silver and sulphuric acid plus state backing. A standalone Western smelter cannot. So the West is already losing furnaces: Glencore has wrapped a rescue package around an Australian smelter and Vedanta has shut its Zambian operation. The honest caveat is that today the scarce physical input is concentrate, not furnaces, and negative charges are the signature of a smelter glut fighting over ore. Our view is that this glut culls capacity asymmetrically, killing the uneconomic ex-China furnaces while China and state-backed Indian and Indonesian builds absorb the survivors, so by the early 2030s qualified non-China conversion, not extraction, becomes the gate. Wood Mackenzie already puts the cost of routing around China at 85 billion dollars.				
P2	By 2033 a major US transmission or large-load interconnection program (an SDA-scale line, a multi-GW datacenter cluster, or a regional RTO buildout) publicly names a shortage of qualified protection-and-control / relay /	Settings-grade substation protection-and-control / relay engineers and journeyman transmission linemen: a fixed, slow-to	78%	46%	2033-12-31

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	substation engineers OR journeyman transmission linemen as the schedule-driving energization constraint, ranked ahead of transformer or switchgear lead time. Concretely: at least one FERC-jurisdictional utility or hyperscaler discloses an energization slip attributed to engineering or craft-labor headcount (not gear), AND all-in real compensation for licensed substation P&C engineers rises more than 50 percent over 2025 as firms bid for a fixed pool.				
P3	D-T fusion needs tritium, and tritium self-sufficiency needs a breeding blanket of lithium enriched to 30 to 90 percent Li-6. Natural lithium is only 7.5 percent Li-6, so every commercial reactor needs an enrichment step. The single industrial method ever scaled, the COLEX column-exchange process, runs on liquid mercury, which is exactly what the 2013 Minamata Convention exists to phase out. The US banned COLEX in 1963 and has zero active Li-6 enrichment capacity today. Russia and China are the only operators, and China is widely believed to buy from Russia rather than run its own line. A pilot plant needs tens of tonnes per year of enriched Li-6, commercial fusion needs hundreds, and there is no liquid Western market for it. So a Western fusion company that solves confinement and	Western-legal enriched lithium-6 (30 to 90 percent Li-6) separation capacity, which is effectively zero. The only scaled	78%	42%	2034-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	reaches D-T fueling around 2032 to 2035 hits a wall: it cannot legally build the only enrichment process proven at scale, and the incumbents who can are geopolitical rivals. The rent moves off the tokamak onto a few hundred kilograms a year of an isotope nobody in the West currently makes. Clean routes (mercury-free electrochemical, chromatographic) exist but are pre-commercial, and their own scale-up estimates run about a decade. The catalyst is mechanical: the fueling milestone arrives on a fixed physics timeline, the enrichment plant does not exist, and a Russian or Chinese export license sits in between.				
P4	The shortage everyone funds is at the factory. Nexans and Prysmian plants are sold out for years and three firms own most of HVDC, so capital and headlines pour into new cable lines. Plants are the fast layer: you can pour concrete and stand up a line in two to three years, and a wave of expansions lands 2029 to 2031. The rent then jumps one node downstream to the thing that actually buries cable in the seabed. The global cable-lay fleet is small (roughly 37 vessels in 2025, projected to about 48 by 2030), a DP3 newbuild takes three to four years and over 200 million dollars, and the same hulls are fought over by HVDC, offshore wind, telecom subsea, and	The certified offshore high-voltage jointing crew is the deepest, least-priced lock; the cable-lay/offshore-construction	70%	46%	2032-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	platform electrification at once. The truly inelastic input is not the hull. It is the certified offshore HV jointer who splices and terminates each cable: a four-person team can take 18 hours per joint, the trade needs a multi-year apprenticeship, the senior cohort is retiring, and there is no automation substitute. When plant capacity is solved, the bottleneck does not vanish. It relocates to the deck of a ship and the hands of a few thousand people.				

Case = how strong we think the thesis is. Our call = the probability we put on the exact, dated version landing by the deadline, and the number we are graded on. The gap between them is the cost of pinning a date and a hard threshold, not a lack of conviction.

P1 • By 2033 the binding constraint on Western copper has moved off the mine onto qualified non-China smelting and refining capacity. A sustained zero-to-negative treatment-charge regime forces ex-China smelters into structural contraction through the late 2020s, with at least two further Western or allied smelter closures or curtailments and a formal allied policy intervention to protect what smelting remains.

P1

The boom: We think the copper debate is aimed at the wrong end of the chain. The market fixates on the mine: too few new deposits, decade-long permitting, an 80-percent-of-needs-by-2030 gap. That is real, it is also consensus, and it is priced. The quieter fact is that China runs about 97 percent of global copper smelting and refining capacity, and the economics of conversion now punish everyone else. The treatment and refining charge, the fee a smelter earns to turn concentrate into metal, settled at zero per tonne on the 2026 annual benchmark, the lowest ever agreed, and the spot charge has gone deeply negative, hitting roughly minus 78 dollars a tonne by April 2026. A negative charge means a smelter pays for the right to operate. Chinese smelters survive on byproduct gold, silver and sulphuric acid plus state backing. A standalone Western smelter cannot. So the West is already losing furnaces: Glencore has wrapped a rescue package around an Australian smelter and Vedanta has shut its Zambian operation. The honest caveat is that today the scarce physical input is concentrate, not furnaces, and negative charges are the signature of a smelter glut fighting over ore. Our view is that this glut culls capacity asymmetrically, killing the uneconomic ex-China furnaces while China and state-backed Indian and Indonesian builds absorb the survivors, so by the early 2030s qualified non-China conversion, not extraction, becomes the gate. Wood Mackenzie already puts the cost of routing around China at 85 billion dollars. • Domain: materials

STRUCTURAL CASE

72%

OUR CALL, DATED

55%

RESOLVES

2033-12-31

Smelters are multi-billion-dollar, multi-year, environmentally hard-to-permit assets that are not rebuilt once shut, so closures are close to irreversible on this horizon. With charges at or below zero, ex-China primary smelting is cash-negative and depends on byproduct credits and balance-sheet endurance that Chinese and state-backed plants have and standalone Western plants do not. That makes the contraction of ex-China capacity genuinely inelastic on the downside. The weaker link is how we frame the input: the current scarce input is concentrate, and negative charges signal a smelter glut, not refining scarcity. The thesis survives only through the asymmetric-culling mechanism, where the glut removes uneconomic ex-China furnaces faster than it removes Chinese or new state-backed Asian furnaces, leaving allied conversion as the constraint even when global furnaces are ample.

WHY IT IS NOT PRICED YET

Genuinely ahead of consensus at the equity-pricing and mainstream-narrative level, where copper is still a mine-and-demand story. Only partly ahead of the specialists: the smelting chokepoint, China's 97 percent share, the negative-charge regime, the Wood Mackenzie 85bn gap and the Columbia CGEP call to protect Western smelting are all in print. The specific, datable claim that remains under-priced is that the late-2020s capacity shakeout resolves asymmetrically and leaves qualified non-China refining as the binding gate by the early 2030s.

WHERE THE PRICE SITS TODAY

Equity and broad energy-transition narratives still trade copper as a mine-supply and demand-growth story (S&P Global and ICSG deficit headlines, permitting, grades). The smelting and refining chokepoint is well covered in specialist and trade channels (Fastmarkets, CRU, Wood Mackenzie, Columbia CGEP) and shows up in TC/RC indices, which is exactly where smelter stress is already priced. The narrow claim, that asymmetric capacity culling makes qualified non-China conversion the binding gate by the early 2030s, is not in equities or the mainstream transition narrative, but it is no longer obscure among policy and trade specialists. Mid-tier on the price channel, not deeply hidden.

THE BINDING CONSTRAINT

Qualified non-China copper smelting and refining capacity. The dated catalyst is the zero-to-negative treatment-charge regime (2026 benchmark at zero per tonne, spot near minus 78 per tonne) persisting and forcing ex-China smelter curtailments and closures through the late 2020s, leaving allied refining as the gating node by the early 2030s.

WHAT WE ARE WATCHING

Track now: (1) the annual copper smelter TC/RC benchmark and the spot TC/RC index (zero or negative equals stress; 2026 benchmark confirmed at zero per tonne, spot near minus 78 per tonne April 2026); (2) announced curtailments, closures, or state bailouts of smelters in the US, EU, Japan, Korea, Australia, Chile, Zambia and Canada (baseline already includes the Glencore Australia rescue and Vedanta Zambia shutdown); (3) ex-China share of global refined-copper output (China about 60 percent of output on about 97 percent of capacity) and ramp rates at Adani Mundra and Freeport Manyar; (4) any US, EU or allied policy action explicitly framed to protect or rebuild domestic smelting capacity, against the Columbia CGEP proposal as a marker.

WHAT WOULD PROVE US WRONG

By 2033-12-31, treatment and refining charges have recovered durably to positive double-digit dollars per tonne AND ex-China smelting capacity is stable or growing with no net additional Western or allied smelter closures or curtailments beyond those already announced by mid-2026 and no allied policy intervention specifically framed to protect smelting; OR new non-China refining capacity (notably Adani Mundra at 0.5 to 1.0 Mt and Freeport Manyar at 0.48 Mt scaling to design rates, plus further Indian or Indonesian builds) reaches utilization sufficient that allied refining stops being the binding gate on Western copper. Note: because negative charges are driven by a concentrate shortage, a durable concentrate-supply recovery that lifts charges back positive without any Western closures also kills the thesis.

HOW WE TRIED TO BREAK IT

The strongest argument against us is causal. Negative TC/RCs are caused by too much smelting capacity chasing too little concentrate, so the physically binding input right now is ore, not furnaces. If concentrate supply recovers (new mine output, the ICSG-flagged swing back toward concentrate balance), charges rise, Western smelters stop bleeding, and the thesis collapses without any of the predicted closures. Second, the glut is partly self-correcting inside China itself: the CSPT plans to cut Chinese primary smelting capacity by over 10 percent in 2026, which relieves global oversupply and props charges back up, weakening the asymmetric-culling mechanism. Third, large new ex-China capacity is being commissioned now, not retired: Adani Mundra (0.5 to 1.0 Mt) and Freeport Manyar (0.48 Mt) are ramping, so non-China refining is expanding, which cuts directly against refining becoming the Western gate. The thesis survives these because the

closures are not hypothetical (Glencore Australia rescue, Vedanta Zambia shutdown are already on the board), because the new Indian and Indonesian capacity is starved of concentrate and running far below nameplate, and because reshoring allied conversion to qualified, permittable, non-China plants is a distinct and harder problem than adding state-backed Asian tonnes. But the case against is strong enough to cap the odds on the exact dated version near a coin flip. The eight-year horizon, the concentrate-versus-furnace ambiguity, and the live counter-trend of new Asian capacity all tax the exact resolution.

Why we are making the call We promote this, with real humility on the dated call. The structural case is strong (conviction 0.72): China's roughly 97 percent capacity share, a zero benchmark and deeply negative spot charges, irreversible smelter assets, real ex-China closures already on the board, and an active allied policy debate. The exact dated version is taxed by a genuine causal ambiguity (the live binding input is concentrate, not refining) plus two counter-trends (Chinese self-cuts that lift charges, and new Adani and Freeport capacity that expands non-China refining), so we put 55 percent on it, near a coin flip. The two-closures-plus-policy-intervention sub-condition is the part most likely to resolve YES; the cleaner "refining is THE binding Western gate by the early 2030s" framing is where the risk concentrates. Inelastic on the downside, ahead of consensus at the pricing level, survives the case against on the narrow capacity-culling mechanism. Keep it, track the TC/RC index and ex-China utilization as the live tells, and treat a durable positive-charge recovery without Western closures as the clean kill.

P2 · The grid and AI-datacenter buildout gets gated through the 2030s not by transformers, steel, or switchgear, but by the certified people who specify, set, and energize substations. Settings-grade protection-and-control relay engineers and journeyman transmission linemen cannot be minted fast enough, and the pipeline to make more of them is itself broken.

The boom: By 2033 a major US transmission or large-load interconnection program (an SDA-scale line, a multi-GW datacenter cluster, or a regional RTO buildout) publicly names a shortage of qualified protection-and-control / relay / substation engineers OR journeyman transmission linemen as the schedule-driving energization constraint, ranked ahead of transformer or switchgear lead time. Concretely: at least one FERC-jurisdictional utility or hyperscaler discloses an energization slip attributed to engineering or craft-labor headcount (not gear), AND all-in real compensation for licensed substation P&C engineers rises more than 50 percent over 2025 as firms bid for a fixed pool. · Domain: energy / human-capital

STRUCTURAL CASE

78%

OUR CALL, DATED

46%

RESOLVES

2033-12-31

This is a demographic and pipeline lock, not a price signal. Nearly half the power-sector workforce is retirement-eligible by 2030 and grid-trade retirements outrun new entrants about 1.4 to 1 (confirmed live). The feeder pipeline is physically capped: fewer than 30 US power-engineering programs after two decades of attrition, retired faculty left unreplaced, and a journeyman lineman or settings-grade relay engineer takes 4 to 10 years that capital cannot compress. Goldman puts the need above 750,000 new power workers by 2030 (~386,000 in T&D including retirements) while the apprenticeship run-rate sits near 45,000/yr against the ~65,000/yr needed. Every transformer, GOES, and switchgear call in the map sits downstream of the humans who specify, wire, and commission the gear.

WHY IT IS NOT PRICED YET

As of mid-2026 the market prices the HARDWARE bottleneck: transformer lead times at 160-plus weeks, switchgear, and China-exposed parts are the named schedule drivers, and labor is treated as a secondary, hireable input that clears at some wage. The broad workforce-shortage narrative is well covered (Goldman 750k by 2030, Utility Dive, POWER), and PE is already buying the PLANNING / interconnection-study bottleneck (Littlejohn-GDS, March 2026). What is not yet in project timelines or sell-side models is the narrower claim: the specific certified P&C-settings and journeyman-lineman pool is demographically fixed and its refill pipeline (under 30 university programs, unreplaced faculty, multi-year apprenticeships) is broken, so the binding constraint moves from the gear to the people and stays there through the 2030s.

WHERE THE PRICE SITS TODAY

Live anchor: substation P&C engineer comp sits near a 96k median (May 2026), with senior/specialist roles 130-170k. No runaway real-wage explosion yet, so the >50% real-comp condition is genuinely unrealized rather than priced. Transformer lead times (the priced rival constraint) ran 140 weeks in 2023, about 150 in 2025, and 160-plus in 2026. PE capital has entered the planning-consulting layer (Littlejohn-GDS) but not the craft/settings labor pool itself, which is still treated as commodity hire.

THE BINDING CONSTRAINT

Settings-grade substation protection-and-control / relay engineers and journeyman transmission linemen: a fixed, slow-to-replenish certified pool fed by fewer than 30 surviving US power-engineering university programs and 4-to-10-year apprenticeships, with grid-trade retirements outrunning new entrants about 1.4 to 1.

WHAT WE ARE WATCHING

Track now: (1) median advertised comp and time-to-fill for substation P&C / relay-protection engineer roles (live baseline ~96k median, 130-170k senior, May 2026); (2) annual US power-engineering degree conferrals and the count of accredited power-engineering programs; (3) IBEW/utility lineman apprenticeship slots filled vs the ~65k/yr needed run-rate (45k in 2024); (4) any FERC interconnection-queue or large-load filing naming engineering or craft headcount, not gear, as the delay cause; (5) the share of P&C settings roles posted as remote/offshore-eligible (the live elasticity leak to watch).

WHAT WOULD PROVE US WRONG

By 2033 no major US grid or large-load program names engineering or craft-labor headcount as the schedule-driving energization constraint (gear lead times still dominate), OR real comp for substation P&C engineers has risen less than 50 percent over 2025 because the pool cleared via remote-engineering / offshored relay-settings work, automated settings-and-commissioning tooling, or a graduate-and-apprentice surge that closes the 1.4:1 gap before 2030. Any one of these breaks the lock.

HOW WE TRIED TO BREAK IT

We tried to kill it three ways. (1) Already priced: the workforce shortage is heavily covered and PE is rolling up power-engineering consultancies, so the THEME is priced. But the priced rival is gear (transformers 160+ weeks, named as THE schedule driver; labor called secondary), and the PE money is going to planning/interconnection-study capacity, not the narrow P&C-settings and lineman pool. The specific labor-as-named-constraint shift is not in project timelines or equity models. Survives, narrowly. (2) Elasticity: the lineman leg is genuinely inelastic (physical, 4-10 yr, capped apprenticeships). The ENGINEER leg has live leaks: relay-settings work is increasingly posted remote and is offshore-eligible, and settings/commissioning automation is advancing. This is the real weakness, and it is why the engineer half of the call is not a layup. (3) Dated-call tax: the boom is conjunctive (named-constraint AND >50% real comp) by a fixed 2033 date, and current comp shows no runaway. The structural case is strong; the exact dated version is a near coin flip because of the offshoring/automation release valve and the demanding comp threshold.

Why we are making the call We promote it because it sits ahead of consensus (the priced constraint is hardware; labor is treated as secondary and hireable), it is genuinely inelastic on at least the lineman leg with a physically broken refill pipeline, and it survives the case against on the narrow needle even though the broad theme is covered. Our conviction in the thesis is high (0.78): the demographic and pipeline lock is real and confirmed live. The dated call stays honest at 0.46, well below that conviction, because resolution requires BOTH a named labor-not-gear schedule slip AND a >50% real comp jump by 2033, while remote/offshored relay-settings work and settings-automation are live elasticity leaks on the engineer side and current comp shows no runaway. The single best forward signal to watch is the

remote/offshore share of P&C settings postings: if that climbs, the engineer leg deflates and only the lineman leg carries the lock.

P3 · Lithium-6 enrichment, not the reactor, is the licensed chokepoint of the Western fusion fuel cycle. By 2034 a Western D-T pilot publicly names enriched Li-6 supply as the binding constraint, because the only scaled route (mercury COLEX) is banned under Minamata and live capacity sits only in Russia and China.

The boom: D-T fusion needs tritium, and tritium self-sufficiency needs a breeding blanket of lithium enriched to 30 to 90 percent Li-6. Natural lithium is only 7.5 percent Li-6, so every commercial reactor needs an enrichment step. The single industrial method ever scaled, the COLEX column-exchange process, runs on liquid mercury, which is exactly what the 2013 Minamata Convention exists to phase out. The US banned COLEX in 1963 and has zero active Li-6 enrichment capacity today. Russia and China are the only operators, and China is widely believed to buy from Russia rather than run its own line. A pilot plant needs tens of tonnes per year of enriched Li-6, commercial fusion needs hundreds, and there is no liquid Western market for it. So a Western fusion company that solves confinement and reaches D-T fueling around 2032 to 2035 hits a wall: it cannot legally build the only enrichment process proven at scale, and the incumbents who can are geopolitical rivals. The rent moves off the tokamak onto a few hundred kilograms a year of an isotope nobody in the West currently makes. Clean routes (mercury-free electrochemical, chromatographic) exist but are pre-commercial, and their own scale-up estimates run about a decade. The catalyst is mechanical: the fueling milestone arrives on a fixed physics timeline, the enrichment plant does not exist, and a Russian or Chinese export license sits in between. · Domain: energy

STRUCTURAL CASE

78%

OUR CALL, DATED

42%

RESOLVES

2034-12-31

Consensus prices fusion's science risk (net energy, magnets, materials) and treats fuel as solved because deuterium is abundant in seawater. Almost nobody outside a thin fuel-cycle literature prices that the Li-6 enrichment step is both physically required for tritium self-sufficiency AND legally blocked in the West by a mercury treaty, with the only working capacity held by Russia and China. The existing FUTURE_MAP covers He-3 and dilution fridges for quantum and HALEU enrichment for fission SMRs, but Li-6 enrichment for the fusion fuel cycle is absent. Different isotope, different process (COLEX/mercury vs gas centrifuge), different treaty hook (Minamata, not non-proliferation), different demand driver (private fusion D-T fueling). The seam is the intersection: a banned process plus an adversary monopoly plus a fixed physics deadline.

WHY IT IS NOT PRICED YET

Fusion coverage and capital concentrate on confinement physics, magnets, and net-energy timelines; the fuel cycle is assumed solved via abundant deuterium. The Li-6/Minamata/COLEX monopoly chain is confined to a narrow fuel-cycle literature (a 2025 Joule paper, FAS/policy notes) with no equity or commodity market pricing it and no slot in the disruptive-foresight map. The substitute programs that could break it are public but pre-commercial, so the constraint is acknowledged in the lab while staying unpriced in the market.

WHERE THE PRICE SITS TODAY

No liquid Western market and no commodity or equity pricing. Enriched Li-6 trades only in thin specialty supply quoted around \$50,000/kg, against a paper-estimated at-scale floor near \$1,000 to \$2,000/kg, and the spread itself reflects the absence of real capacity. Recent 2026 DOE enrichment money (\$2.7B) went to

uranium (HALEU/LEU), not lithium, so the public-capital signal has not yet flowed to the named constraint. The market channel is genuinely unpriced, which supports PROMOTE.

THE BINDING CONSTRAINT

Western-legal enriched lithium-6 (30 to 90 percent Li-6) separation capacity, which is effectively zero. The only scaled route, the mercury-based COLEX process, is banned for new use under the Minamata Convention, and the clean substitutes (electrochemical V2O5 insertion, displacement chromatography) are pre-commercial. Live industrial capacity exists only in Russia and China. Specialty supply quoted near \$50,000/kg; the paper's at-scale economic floor is roughly \$1,000 to \$2,000/kg.

WHAT WE ARE WATCHING

Track: (1) public statements or SEC/investor disclosures from private fusion firms (CFS, TAE, Tokamak Energy, Type One, Helion, Zap) naming Li-6 or enriched-lithium supply as a fuel-cycle risk; (2) DOE/NNSA solicitations or awards for domestic non-mercury Li-6 enrichment, and the operational readiness level of named programs (Precision Periodic DOE-grant line announced Sep 2025, ORNL/PSI/HBKU electrochemical V2O5 route published Mar 2025) - watch for movement from lab/pilot to qualified tonnes-per-year output; (3) any Russian or Chinese export-licensing notice covering enriched lithium isotopes (HS 284530); (4) emergence of any spot/contract reference price for enriched Li-6 metal, currently effectively no liquid Western market (specialty supply near \$50k/kg).

WHAT WOULD PROVE US WRONG

By 2034-12-31 no Western fusion or breeding-blanket program publicly cites enriched lithium-6 supply as a binding constraint, AND either (a) a non-mercury Li-6 enrichment line (electrochemical, chromatographic, or laser) reaches qualified commercial-scale output (order tonnes per year) in the US or allied bloc, or (b) D-T fusion fueling milestones slip past 2035 so the demand never binds in window. Also killed if a credible blanket design demonstrates tritium self-sufficiency (TBR above 1) using natural unenriched lithium at scale.

HOW WE TRIED TO BREAK IT

The hardest argument against us is the candidate's own kill line (b): the clean substitutes are not idle, they are funded and advancing inside the resolution window. A March 2025 mercury-free electrochemical route (selective Li-6 insertion in 1D tunnel-structured V2O5) matched COLEX performance in the lab (about 5.7% enrichment per cycle, 45 cycles to 90% Li-6), and in September 2025 Precision Periodic announced a DOE-grant-backed scalable Li-6/Li-7 enrichment process (displacement chromatography, simulated moving bed). Both are pre-commercial today, but their stated scale-up horizon is roughly a decade, which overlaps 2034. So a US/allied qualified line could plausibly exist before the deadline, tripping the kill. The second live counter is timing on the demand side: most private fusion DT-fueling milestones already slip, and if fueling slips past 2035 the demand never binds in window. The thesis still survives because BOTH escape routes must fire on time and at scale for the dated call to fail, and neither has yet cleared lab/pilot to qualified commercial output. Meanwhile the market channel is genuinely unpriced (no liquid Western market, no equity coverage, no commodity reference price) and the input is genuinely inelastic (zero Western legal capacity, controlled substance, banned primary process, adversary monopoly). What it survives on: the mechanism is now peer-reviewed (Joule 2025), not speculative, yet capital and coverage are still pointed at confinement physics and treat fuel as solved via abundant deuterium.

Why we are making the call Our read on the market channel holds. A peer-reviewed 2025 Joule paper now states the mechanism plainly, yet there is no equity coverage, no commodity reference price, and no liquid Western market for enriched Li-6, while fusion capital stays fixed on confinement physics and treats fuel as solved via abundant deuterium. The input is genuinely inelastic near-term: zero Western legal capacity, a banned primary process, a controlled substance, and an adversary monopoly. It survives the case against because the two escape routes (a qualified clean line before 2034, or fusion fueling slipping out of window) are real but neither has cleared lab/pilot to qualified commercial output, and both would have to fire on time. Our conviction in the thesis is high because the structural chain is now confirmed, not speculative. The dated call sits near a coin flip because the exact dated version carries a heavy timing-and-substitution tax: the substitute scale-up clock and the private-fusion fueling clock are both running inside the same window the call must resolve in.

P4 · By 2032 the binding grid-connection constraint has moved downstream from cable manufacturing to marine installation, with the certified high-voltage offshore jointing crew as the deepest lock. At least three sanctioned HVDC links slip commissioning citing installation-vessel or jointing-crew availability (distinct from cable supply), and installation-slot lead times for HVDC campaigns exceed cable-plant lead times.

The boom: The shortage everyone funds is at the factory. Nexans and Prysmian plants are sold out for years and three firms own most of HVDC, so capital and headlines pour into new cable lines. Plants are the fast layer: you can pour concrete and stand up a line in two to three years, and a wave of expansions lands 2029 to 2031. The rent then jumps one node downstream to the thing that actually buries cable in the seabed. The global cable-lay fleet is small (roughly 37 vessels in 2025, projected to about 48 by 2030), a DP3 newbuild takes three to four years and over 200 million dollars, and the same hulls are fought over by HVDC, offshore wind, telecom subsea, and platform electrification at once. The truly inelastic input is not the hull. It is the certified offshore HV jointer who splices and terminates each cable: a four-person team can take 18 hours per joint, the trade needs a multi-year apprenticeship, the senior cohort is retiring, and there is no automation substitute. When plant capacity is solved, the bottleneck does not vanish. It relocates to the deck of a ship and the hands of a few thousand people. · Domain: energy

STRUCTURAL CASE

70%

OUR CALL, DATED

46%

RESOLVES

2032-12-31

Cable plants are the fast, fundable layer (two to three years to stand up a line). The installation fleet is slow (three to four years and over 200 million dollars per DP3 newbuild) but is being actively addressed by public hull orders. The certified offshore HV jointer is the deepest inelastic input: a multi-year apprenticeship, a retiring senior cohort, about 18 hours per joint for a four-person team, and no automation substitute today. Hull additions and throughput gains do not mint jointers, so the labor node stays inelastic even as steel is added.

WHY IT IS NOT PRICED YET

The verdict splits by layer. The cable-MANUFACTURING constraint is fully priced. The cable-VESSEL layer is now partly priced, not obscure: Boskalis, Jan De Nul, Nexans and DEME are publicly ordering hulls, sell-side and market-research coverage names vessel-capacity bottlenecks outright, and equities react to fleet additions. The genuinely UNPRICED layer is the certified jointer labor pool, discussed only in recruiter and trade press as a contributing factor and carrying no analyst or equity coverage as a structural gate. The thesis survives on the labor-pool node, not the vessel node.

WHERE THE PRICE SITS TODAY

There is no clean tradable spot price for jointer labor. The live anchors are charter/day-rate inflation for high-spec CLVs (rising, but already reflected in installer order activity and valuations) and export-cable award-to-installation lead time, which has roughly doubled since 2018 to over 800 days on array cabling. Vessel-side rents are being competed for openly through newbuild orders, so the vessel layer is closer to priced than the candidate claims. The jointer-pool scarcity has no price channel anyone trades, which is exactly why we read it as the live edge.

THE BINDING CONSTRAINT

The certified offshore high-voltage jointing crew is the deepest, least-priced lock; the cable-lay/offshore-construction vessel fleet is the second lock. Catalyst: the 2029-2031 cable-plant expansion wave (Nexans, Prysmian, NKT, Hengtong) removes the plant gate and exposes installation as the residual constraint.

WHAT WE ARE WATCHING

(1) Global cable-lay/offshore-construction vessel orderbook and average HVDC installation-slot lead time quoted by Boskalis, DEME, Jan De Nul, Prysmian and Nexans marine units, versus cable-plant lead time. (2) Day rates and wait times for HVDC cable installation campaigns, and any HVDC or offshore-wind project status report naming vessel or jointing-crew availability as a schedule driver. (3) Headcount and training-pipeline throughput of certified offshore HV jointers (NKT/Prysmian/Nexans certifications). Today plant backlog dominates the narrative; jointer-crew scarcity appears only in trade press.

WHAT WOULD PROVE US WRONG

By 2032-12-31, HVDC cable installation-slot lead times remain shorter than cable-plant lead times AND no sanctioned HVDC or major offshore-wind project cites installation-vessel or jointing-crew availability (distinct from cable supply) as a schedule-driving constraint; or fleet newbuilds plus automated/robotic jointing scale enough that installation throughput stops binding.

HOW WE TRIED TO BREAK IT

Two real counters. (1) Already priced on the vessel node: installers are visibly ordering hulls and analysts name the bottleneck, so betting "vessels gate" captures little rent. The thesis has to lean on the jointer pool, where coverage is absent. (2) Supply is more elastic than implied: the fleet is set to grow about 37 to 48 by 2030 (nine newbuilds by end-2026), dual-lane high-capacity vessels raise per-hull throughput, and jointing automation/robotic termination R&D could ease the labor lock. These elasticities cap our conviction in the thesis below certainty. The thesis survives because hull and throughput gains do not mint jointers, the senior cohort is retiring, and the apprenticeship pipeline cannot compress to match a 2029-2031 demand spike; the labor node stays inelastic even as steel is added.

Why we are making the call We promote it on the labor-pool node, and we stay demote-aware on the vessel node. The mechanism is sound and partly live-confirmed: plants are the fast, fundable layer; vessels are slow but being addressed; certified HV jointers are the deepest inelastic input, with a multi-year, retiring, non-automatable pipeline and no price channel. The structural case is strong (conviction 0.7), but the exact dated version carries a heavy tax: it requires at least three sanctioned HVDC links to slip AND explicitly cite vessel-or-crew (not cable) availability AND installation lead time to exceed plant lead time by 2032, a hard, multi-condition, timing-sensitive bar that the partly-elastic fleet buildout and possible jointing automation can defeat. Hence the dated call sits at 0.46, near a coin flip, well below our conviction in the thesis.

Seeds considered and not promoted

These cleared the supply-side test but did not make the cut, usually because there was no clean way to express the trade or the move was already in the price. We list them because we would rather show our work than bury what we dropped.

SEED	PHYSICAL CASE	WHY NOT PROMOTED
Enriched lithium-6 is the binding gate on commercial D-T fusion (mercury COLEX Minamata-restricted, capacity only in Russia and China, export-controlled as thermonuclear feed); by 2034 a fusion power-demo program publicly names Li-6 enrichment supply as its limiting input while Western qualified output stays under 10 t/yr.	energy-fusion	Near-duplicate of P3 (same Li-6/COLEX/Minamata/adversary-monopoly mechanism, same 2034 window). P3 is the stronger framing: it ties the demand catalyst to named private-fusion D-T fueling milestones and a single fusion-pilot constraint statement, whereas this variant adds a brittle quantitative sub-condition (under 10 t/yr Western output) and its own refute concedes the named frontrunner ICOMAX is itself a mercury-amalgam process, which blurs the needle. Lower odds on the exact dated version (0.34). We kept one Li-6 call to preserve mechanism diversity.
By 2034 a Western D-T pilot names enriched Li-6 supply as the binding constraint (the second Li-6 variant emphasizing ICOMAX/KIT/Kyoto Fusioneering cleaner-amalgam, laser AVLIS, and the natural-unenriched-lithium-blanket kill path).	energy	Third copy of the same Li-6 fusion thesis. Two Li-6 calls in the top 8 would crowd out a distinct mechanism for no added edge. P3 already carries the strongest version of this constraint; we fold this one into P3 and the first runner-up.

Each call is dated, with the line that would prove us wrong fixed the day we make it. We do not move the goalposts: a call is superseded, never quietly edited, and graded in public when its date arrives.